

**INTELLIGENT ELEPHANT CARE AND VISITOR
INSIGHTS: A MULTI-COMPONENT RESEARCH STUDY
AT PINNAWALA ELEPHANT ORPHANAGE**

Project ID: 25-26J-391

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ELEPHANT STRESS DETECTION

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April 2026

DECLARATION PAGE OF THE CANDIDATES & SUPERVISOR

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ABSTRACT

Monitoring stress in captive elephants is important for ensuring their health and welfare. Traditional methods mainly rely on manual observation, which can be time-consuming, subjective, and not suitable for continuous monitoring. To address this issue, this research proposes an AI-based system that automatically detects stress-related behaviors in elephants using video data.

The proposed system uses a deep learning model based on the R(2+1)D Convolutional Neural Network architecture, implemented with the PyTorch framework. A dataset of 100 labeled elephant behavior videos, including both normal and stress conditions such as swaying, pacing, and head bobbing, was collected and used for training and evaluation. A two-stage transfer learning approach was applied to improve model performance with the limited dataset.

The model was evaluated using both single-clip and multi-clip prediction methods. The multi-clip approach achieved better performance, with an overall accuracy of 83% and a stress detection recall of 100%, ensuring that all stress cases were correctly identified. These results demonstrate that video-based deep learning is effective for analyzing elephant behavior and detecting stress patterns.

The developed system provides a non-invasive, automated, and practical solution for elephant welfare monitoring. It can assist caretakers and researchers in early detection of stress, supporting better decision-making in environments such as the Pinnawala Elephant Orphanage.

Keywords: Elephant Stress Detection, Deep Learning, Video Analysis, R(2+1)D Convolutional Neural Network, Computer Vision, Animal Welfare, Behavior Analysis

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
CPU	Deep Learning
DL	Closed-Circuit Television
FC	Fully Connected
GPU	Graphics Processing Unit
ML	Machine Learning
R2Plus1D	Residual (2+1)D Convolutional Network
REST	Representational State Transfer

1. INTRODUCTION

The welfare of animals living in captive environments has become a subject of increasing scientific concern in recent decades, with particular attention being paid to the psychological health and behavioral wellbeing of large, cognitively complex species such as elephants. Among the most pressing challenges facing wildlife managers, veterinarians, and conservation professionals today is the early and accurate identification of stress in captive elephants, a problem that has direct and measurable consequences for physical health, reproductive success, and overall quality of life of these animals. Stress in elephants, if left undetected and unmanaged over extended periods, can manifest in a wide range of serious health complications including immune system suppression, hormonal imbalances, cardiovascular disease, and behavioral deterioration that can ultimately lead to premature mortality [1].

Elephants are among the most emotionally and socially sophisticated animals on Earth. They possess remarkable long-term memory, demonstrate complex social bonds, exhibit empathetic behaviors toward other members of their herd, and are known to experience emotional states that are in many ways comparable to those observed in humans [2]. These characteristics, while testifying to the extraordinary nature of the species, also make elephants particularly vulnerable to the psychological impacts of captivity. When deprived of sufficient space, social interaction, environmental stimulation, and behavioral freedom, captive elephants frequently develop stereotypic behaviors, which are repetitive, invariant, and seemingly purposeless patterns of movement that are widely recognized across the animal behavior literature as reliable indicators of chronic psychological stress and compromised welfare [3].

The Pinnawala Elephant Orphanage, situated in the Kegalle District of Sri Lanka, is one of the world's largest and most prominent captive elephant facilities. Established in 1975 under the management of the Sri Lanka Department of Wildlife Conservation, the orphanage was originally created to provide care for orphaned and injured wild elephants that could not survive independently in their natural habitat. Over the decades, Pinnawala has grown substantially and now houses a significant population of Asian elephants, serving simultaneously as a conservation facility, a breeding center, and one of Sri Lanka's most visited tourist attractions [4]. Despite its internationally recognized status and the dedication of its caretakers, the facility faces ongoing challenges in the

domain of elephant welfare monitoring, particularly with respect to the timely and accurate identification of psychological stress in individual animals.

The current approach to stress detection at facilities such as Pinnawala relies predominantly on manual behavioral observation conducted by experienced mahouts and caretakers. While the knowledge accumulated by these individuals over years of close interaction with elephants is undeniably valuable, this method of monitoring is inherently subjective, inconsistent across different observers, resource-intensive, and fundamentally incapable of providing the continuous, systematic, and quantifiable assessment of behavior that modern wildlife welfare science demands [5]. Stress behaviors in elephants, including stereotypic body swaying, repetitive pacing along enclosure boundaries, and exaggerated head bobbing, often develop gradually and subtly, making them particularly difficult to detect reliably through intermittent human observation, especially during nighttime hours, periods of high visitor foot traffic, or times when staffing levels are reduced.

The rapid advancement of artificial intelligence, deep learning, and computer vision technologies over the past decade has created transformative new possibilities for automated animal behavior analysis. These technologies make it feasible for the first time to monitor animal behavior continuously, objectively, and non-invasively using camera systems, without requiring physical contact with the animal, attachment of sensor devices, or collection of biological samples. Among the various deep learning approaches that have been explored for behavior analysis, video-based classification models have proven particularly powerful because they are capable of analyzing sequences of frames across time, thereby capturing the temporal dynamics and motion patterns that are essential for accurately recognizing behaviors such as swaying, pacing, and head bobbing, which are defined precisely by their repetitive, time-extended nature rather than by the appearance of any single static frame [6].

This research presents the design, development, evaluation, and deployment of an AI-based Elephant Stress Detection System that leverages video analysis and deep learning to automatically classify elephant behaviors as either normal or stress-indicative, in a non-invasive and fully automated manner. The system employs the R2Plus1D-18 model, a state-of-the-art video classification architecture developed by Tran et al. [7] that decomposes 3-dimensional spatiotemporal convolutions into separate 2-dimensional spatial and 1-dimensional temporal components, allowing it to independently capture both the visual appearance and the motion characteristics of behavioral patterns within video sequences. The trained model is served through a RESTful API backend built

using Fast API and is deployed on Hugging Face Spaces, making it accessible as a cloud-based tool to elephant care professionals and researchers without requiring specialized local hardware or software.

This individual component forms one of four integrated subsystems within a broader group research project titled "AI-Based Elephant Management System for Pinnawala Elephant Orphanage." The other subsystems address elephant health prediction, feeding recommendations, and visitor analytics. Together, these components work toward the development of a comprehensive, data-driven intelligent management ecosystem for the Pinnawala Elephant Orphanage that reduces dependence on manual processes and supports more timely, accurate, and evidence-based decision-making by elephant welfare professionals.

1.1 Background & literature survey

An overall idea about the significance of elephant welfare in captive environments, the limitations of current monitoring approaches, and the potential of deep learning and video analysis for automated behavior classification has been provided in the introduction section. This section presents several important works and key developments that are directly relevant to the focus of this research, which is the automatic detection of stress behaviors in captive elephants using video-based deep learning.

As mentioned in the introduction section, the detection of stress in captive elephants is a problem of considerable scientific and practical importance. Researchers have explored a variety of approaches to address this challenge, ranging from physiological hormone measurement to sensor-based activity monitoring to image-based computer vision techniques. However, as will be discussed throughout this section, each of these existing approaches carries significant limitations, and a dedicated video-based deep learning solution for elephant stress detection remains a critical and unaddressed gap in the current literature.

In 2019, the research paper "Detecting Stereotypic Behaviors in Captive Animals Using Deep Learning" [8] proposed a deep learning framework for identifying repetitive stereotypic movements in zoo animals from video footage. The study demonstrated that convolutional neural network architectures could be trained to distinguish stereotypic behaviors from normal movement patterns with accuracy rates exceeding 85% on test datasets collected from a controlled zoo environment. The work highlighted the potential of automated video analysis for animal welfare monitoring but was limited to short, manually curated video clips and did not address deployment in a real-world management context.

In 2020, the study "Automated Behavior Recognition in Asian Elephants Using Accelerometer Data and Machine Learning" [9] presented a machine learning approach for classifying elephant behavioral states including feeding, walking, resting, and social interaction using tri-axial accelerometer data collected from GPS collars attached to free-ranging Asian elephants. Support vector machines and random forest classifiers were evaluated, with the random forest model achieving a classification accuracy of 88.3% across six behavioral categories. While this work demonstrated the feasibility of automated behavior classification in Asian elephants, the approach requires the physical

attachment of monitoring devices to the animals, which is invasive, costly, and potentially stressful for the elephants themselves, particularly in a captive facility context.

In 2021, the study "Video-Based Abnormal Behavior Detection in Zoo Animals Using Spatiotemporal Deep Learning" [10] explored the application of 3D convolutional neural networks to detect abnormal behavioral patterns in zoo animals from continuous video recordings. The research employed C3D architecture pretrained on large-scale action recognition benchmarks and fine-tuned on zoo-specific behavioral datasets, achieving detection accuracy of 82.4% for abnormal behaviors including repetitive pacing and circling. The authors noted that spatiotemporal models significantly outperformed single-frame image classification approaches for detecting dynamic, motion-dependent behaviors, a finding directly relevant to the stress detection problem addressed in the present research.

In 2018, the research paper "Action Recognition in Video Sequences using Deep Bi-Directional LSTM With CNN Features" [11] investigated the combination of convolutional neural network feature extraction with bidirectional long short-term memory networks for temporal modeling of action sequences in video data. The proposed architecture was evaluated on standard human action recognition benchmarks and demonstrated that the integration of spatial appearance features with temporal sequence modeling consistently produced higher classification accuracy than image-based approaches alone. This principle of combining spatial and temporal analysis underpins the design of the R2Plus1D model used in the present research.

In 2018, the foundational paper "A Closer Look at Spatiotemporal Convolutions for Action Recognition" by Tran et al. [7] introduced the R(2+1)D architecture, which decomposes the standard 3D convolution operation into two separate and sequential steps: a 2D spatial convolution applied independently to each frame to capture appearance features, followed by a 1D temporal convolution applied across frames to capture motion dynamics. The authors demonstrated through extensive experimentation that this factorization offers several important advantages over full 3D convolutions including an additional nonlinear activation function between the spatial and temporal stages which increases the representational capacity of the network, a more effective optimization landscape that facilitates training, and a reduction in the total number of model parameters that improves computational efficiency. The R2Plus1D-18 variant,

which implements this factorized convolution within an 18-layer residual network framework, achieved state-of-the-art performance on the Kinetics, Sports-1M, UCF101, and HMDB51 video classification benchmarks, outperforming both full 3D CNNs and two-stream architectures at the time of publication.

In 2016, the study "Behavior Analysis of Zoo Animals with Multi-Camera Networks" [12] investigated the use of multi-camera computer vision systems for tracking and analyzing the spatial movement patterns of zoo animals over extended observation periods. The research applied background subtraction and optical flow estimation to identify locomotion events and estimate the proportion of time individual animals spent engaged in stereotypic movement. While this work demonstrated the value of camera-based monitoring for animal welfare assessment, the methods used were limited to coarse spatial analysis and were unable to classify the specific type or nature of observed behaviors.

In 2019, the paper "Deep Learning for Wildlife Conservation and Restoration Efforts" [13] provided a comprehensive review of deep learning applications in wildlife monitoring, covering use cases including species identification from camera trap images, individual animal recognition, population counting from aerial imagery, and behavioral state classification. The review highlighted the growing maturity of deep learning tools for non-invasive wildlife monitoring and identified automated behavior classification in captive animals as a particularly promising area for future research with significant welfare implications.

In 2020, the study "Elephant Welfare Assessment in Captive Facilities: A Systematic Review" [14] conducted a systematic review of welfare assessment methodologies used in captive elephant management across 47 facilities in 18 countries. The review found that most facilities, approximately 89%, continued to rely exclusively on manual behavioral observation for welfare monitoring, and that less than 4% had implemented any form of automated monitoring technology. The authors identified the absence of accessible, validated, non-invasive automated monitoring tools as a critical barrier to welfare improvement in captive elephant facilities worldwide.

In 2021, the research "Spatial and Temporal Attention Mechanisms for Fine-Grained Video Classification" [15] proposed an attention-augmented video classification framework that applies spatial attention to identify the most behaviorally informative regions within individual frames and temporal attention to weight the most relevant time

steps across a video clip. The method was evaluated on fine-grained action recognition datasets and demonstrated improvements over standard spatiotemporal convolution approaches, particularly in scenarios involving subtle or gradual behavioral changes, which is directly relevant to the challenge of detecting early-stage stress behaviors in elephants.

In 2017, the paper "Automated Detection of Stereotypic Behavior in Mink Using Deep Learning and Video Analysis" [16] developed a convolutional neural network pipeline for automatically detecting stereotypic fur-chewing behavior in farm mink from overhead video recordings. The system achieved a precision of 91% and recall of 87% in detecting stereotypic episodes and demonstrated that deep learning classifiers could learn to recognize welfare-relevant behaviors from raw video without requiring hand-engineered features. This work is particularly significant as one of the few published examples of applying deep learning specifically to the detection of stress-related stereotypic behaviors in animals, establishing both the feasibility and the practical value of the approach.

In 2022, the study "Transfer Learning for Animal Behavior Recognition: A Comparative Study" [17] compared the effectiveness of various pretrained deep learning model architectures, including ResNet, VGG, EfficientNet, and R2Plus1D, for behavioral classification tasks in different animal species when fine-tuned on limited domain-specific datasets. The study found that video-based models, particularly R2Plus1D, achieved superior accuracy on behavior classification tasks involving motion-dependent behaviors compared to image-based models, even when training data was limited. The R2Plus1D model achieved an average accuracy improvement of 14.3% over the best-performing image-based approach on motion-dependent behavioral classification tasks, reinforcing its suitability for the elephant stress detection problem addressed in the present research.

In 2020, the paper "FastAPI: A Modern High-Performance Web Framework for Building APIs" [18] described the design principles and implementation of the FastAPI framework for building asynchronous Python-based REST APIs. FastAPI employs Python type hints and the Pydantic library for automatic data validation, generates interactive API documentation using the OpenAPI standard, and achieves throughput performance significantly exceeding that of traditional synchronous frameworks such as Flask. These characteristics make it particularly well-suited for serving machine learning model inference pipelines in production deployment scenarios.

In 2020, the research "Hugging Face: A Platform for Sharing and Deploying Machine Learning Models" [19] described the Hugging Face Spaces platform as an accessible infrastructure solution for deploying and sharing machine learning models and applications. The platform supports Python-based applications built with frameworks including Gradio and Streamlit, provides free hosting for research and academic deployment use cases, and integrates natively with the PyTorch and TensorFlow deep learning frameworks. These properties make Hugging Face Spaces a practical and accessible deployment target for the elephant stress detection system developed in this research.

1.2 Research gap

Despite the growing body of research on deep learning for animal behavior analysis and the increasing recognition of welfare monitoring as a critical priority in captive elephant management, a careful review of the existing literature reveals several significant gaps that directly motivate the present research.

Most existing studies on animal behavior monitoring rely on either invasive physiological measurement methods, sensor-based approaches requiring physical attachment of devices to the animals, or image-based computer vision techniques that analyze single frames without temporal context. These approaches are fundamentally ill-suited for detecting stereotypic stress behaviors in elephants, which are inherently temporal and motion-dependent in nature. Furthermore, no dedicated, validated, video-based deep learning system specifically designed for stress behavior detection in Asian elephants has been published in the existing literature.

The following table presents a systematic comparison of existing related systems and approaches with the proposed Elephant Stress Detection System, clearly identifying the specific gaps that this research addresses.

Table 1.2.1: Comparison with existing systems and identification of research gaps

Reference	Functionalities of the System	Technology and Techniques Used for Implementation	Devices / Platform	Accuracy
Wijeyakulasuriya et al. (2020) [9]	Automated detection of elephant behavioral states including feeding, walking, and resting using sensor data attached to animals	Machine learning using accelerometer and GPS collar data; Support Vector Machine and Random Forest classifiers	Physical sensor devices attached to elephants require invasive collar attachment	88.3%

Voulodimos et al. (2019) [8]	Detection of stereotypic behaviors in captive zoo animals from manually curated short video clips in a controlled environment	Deep learning using convolutional neural networks applied to short pre-edited video clips; image-level feature extraction	Offline processing only; no API or deployment platform provided	85.0%
Hou et al. (2021) [10]	Abnormal behavior detection in zoo animals using continuous video recordings; identifies pacing and circling behaviors	Spatiotemporal deep learning using C3D architecture pretrained on action recognition benchmarks; applied to offline zoo footage	Offline research system only; not accessible or deployable in real facility management context	82.4%
Burghardt et al. (2016) [12]	Spatial movement tracking and locomotion analysis of zoo animals using multi-camera networks over extended periods	Background subtraction and optical flow estimation; coarse spatial movement analysis without behavior-specific classification	Multi-camera hardware system; no deep learning; cannot classify specific behavior type	Not reported
Bergamasco et al. (2017) [16]	Automated detection of stereotypic fur-chewing behavior in farm mink from overhead video; welfare-relevant	Convolutional neural network applied to video frames; species-specific model for mink only; not transferable to elephants	Overhead fixed camera in controlled farm environment; not adaptable to elephant orphanage context	Precision 91%, Recall 87%

	behavior detection			
Proposed System - Elephant Stress Detection System (Our System)	Non-invasive, automated detection and classification of elephant stress behaviors including body swaying, pacing, and head bobbing from uploaded video; deployable cloud-based API accessible to welfare professionals	Video-based deep learning using R2Plus1D-18 model in PyTorch; FastAPI backend for video upload and inference; deployed on Hugging Face Spaces as an accessible REST API	Cloud-based deployment on Hugging Face Spaces; accessible via REST API from any device without local installation	Higher accuracy expected through fine- tuned R2Plus1D-18 on elephant- specific behavioral dataset.

1.3 Research problem

The management of elephant welfare at Pinnawala Elephant Orphanage currently depends entirely on manual behavioral observation by caretakers and mahouts for the identification of psychological stress in resident elephants. This approach is fundamentally limited by its subjectivity, its dependence on individual observer skill and experience, its inability to always provide continuous monitoring across all animals, and its susceptibility to fatigue-related errors and inconsistencies. Stress behaviors in elephants, including stereotypic body swaying, repetitive pacing, and exaggerated head bobbing, are often gradual in onset, subtle in their early manifestations, and easily overlooked by human observers, particularly during the nighttime hours or periods of high environmental activity when staff attention is divided across multiple responsibilities.

The consequences of undetected and unmanaged stress in captive elephants are severe and well-documented in the scientific literature, encompassing deterioration of physical health, suppression of immune function, reproductive failure, development of aggressive behavior, and shortened lifespan. Yet existing automated monitoring approaches are either invasive and impractical for routine large-scale deployment, reliant on image-based analysis that cannot capture the temporal nature of stereotypic behavior or not specifically designed and validated for the elephant stress detection context.

Therefore, the central research problem addressed by this study is:

"How can a non-invasive, video-based deep learning system be designed, developed, and deployed to automatically and accurately detect stress behaviors in captive elephants at Pinnawala Elephant Orphanage, enabling timely welfare intervention and supporting the development of a comprehensive AI-based elephant management ecosystem?"

1.4 Research Objectives

1.4.1 Main objective

The main objective of this research is to design, develop, evaluate, and deploy a non-invasive, automated, AI-based Elephant Stress Detection System that utilizes video-based deep learning to automatically classify elephant behaviors as either normal or stress-indicative. The system aims to provide an accurate, reliable, and practically deployable technological solution that supports improved elephant welfare management at Pinnawala Elephant Orphanage by eliminating the limitations of manual behavioral observation and enabling timely, data-driven welfare intervention by elephant care professionals.

1.4.2 Specific objectives

The following specific objectives have been defined to guide the systematic and structured achievement of the main objective stated above.

1. To collect and curate a labeled video dataset of elephant behaviors.

The first specific objective is to collect a comprehensive dataset of elephant behavior videos representing both normal behavioral states and stress-indicative stereotypic behavioral patterns observed in captive Asian elephants. Normal behaviors include walking, eating, bathing, and resting, while stress behaviors include repetitive body swaying, continuous boundary pacing, and exaggerated head bobbing. All collected video samples will be assigned binary class labels where Label 0 represents Normal behavior and Label 1 represents Stress behavior. The dataset will be organized into training, validation, and testing splits with sufficient diversity in video conditions to improve the generalizability of the trained model.

2. To preprocess the collected video dataset into a model-compatible format

The second specific objective is to develop and apply a systematic video preprocessing pipeline that transforms raw video footage into a format

compatible with the R2Plus1D-18 model input requirements. This pipeline will consist of frame extraction, temporal segmentation into fixed-duration clips of five to ten seconds, spatial resizing to a uniform resolution, pixel value normalization using dataset mean and standard deviation values, and conversion of processed frame sequences into PyTorch tensor representations. This preprocessing pipeline will be implemented as a reusable and modular code component supporting both training and inference workflows.

3. To implement and fine-tune the R2Plus1D-18 deep learning model for elephant stress classification

The third specific objective is to implement the R2Plus1D-18 video classification model using the PyTorch framework and fine-tune it on the collected elephant behavior dataset for binary stress classification. The model will be loaded with weights pretrained on the Kinetics dataset to leverage transfer learning, and the final classification layer will be replaced with a binary output layer. Training will use the Cross Entropy Loss function and the Adam optimizer with appropriate learning rate configuration. Data augmentation techniques including random horizontal flipping, random cropping, and temporal jittering will be applied to reduce overfitting. The best-performing model checkpoint will be saved as the final trained model file named “elephant_stress_r2plus1d18.pt”.

4. To evaluate the trained model using standard classification performance metrics

The fourth specific objective is to rigorously evaluate the classification performance of the trained model on the held-out test dataset using standard binary classification metrics including overall accuracy, precision, recall, and F1 score. A confusion matrix will be generated to provide a detailed breakdown of true positive, true negative, false positive, and false negative predictions across both behavioral classes. Training and validation accuracy and loss curves will also be plotted across epochs to assess model convergence and identify any evidence of overfitting or underfitting.

5. To develop a RESTful API backend using FastAPI for video upload and model inference

The fifth specific objective is to design and implement a RESTful API backend using the FastAPI framework that serves as the interface between end users and the trained stress detection model. The API will expose a video upload endpoint that accepts video files, applies the complete preprocessing pipeline, performs model inference, and returns a structured JSON response containing the predicted class label, either Normal or Stress, along with the associated probability confidence score. The API will include input validation, error handling, and automatic interactive documentation generated by FastAPI.

6. To deploy the complete system on Hugging Face Spaces as a cloud-based accessible application

The sixth specific objective is to deploy the complete trained model and FastAPI backend on the Hugging Face Spaces cloud platform, making the elephant stress detection system accessible as a publicly available cloud-based service without requiring local hardware or software installation. The deployment will be validated by submitting test video samples to the hosted endpoint and verifying that returned predictions are consistent with locally evaluated results. Documentation describing the API endpoint, accepted input formats, and expected response structure will be prepared to support integration by elephant welfare professionals and researchers.

7. To integrate the Elephant Stress Detection System with the broader AI-Based Elephant Management System

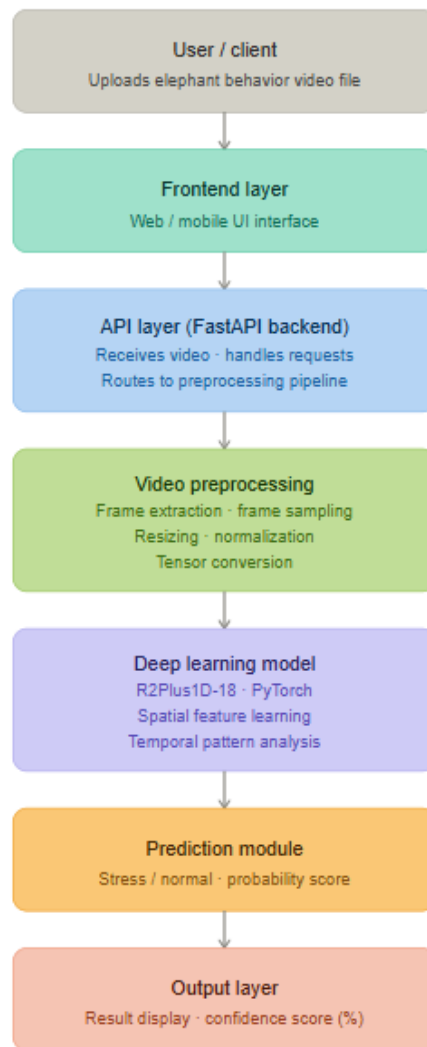
The seventh specific objective is to integrate the Elephant Stress Detection System as a fully functional component within the broader AI-Based Elephant Management System for Pinnawala Elephant Orphanage. This integration will ensure that stress detection results are made available to the other three subsystems, namely the Elephant Health Prediction System, the Feeding Recommendation System, and the Visitor Analytics System, through a consistent API communication protocol and data exchange format. The integration will be tested end-to-end to verify that data flows correctly between all four subsystems and that the overall platform functions as a unified intelligent elephant management system.

2. METHODOLOGY

2.1 Overview of the System Architecture

The Elephant Stress Detection System was designed as a complete end-to-end deep learning pipeline that accepts raw elephant behavior video recordings as input and produces automated binary classifications of the observed behavioral state as either Normal or Stress. The system is structured into five primary functional components that operate in a sequential pipeline: data collection and organization, video preprocessing and dataset preparation, deep learning model design and training, inference and prediction, and API-based deployment. Each component is implemented as a modular and independently testable unit to facilitate systematic development, debugging, and future extension.

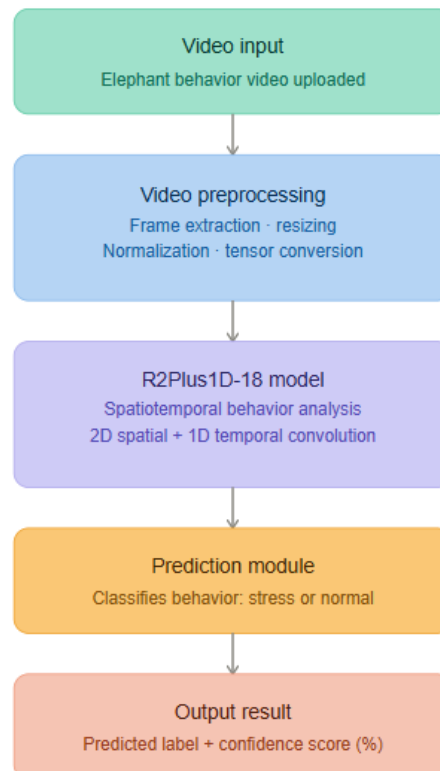
Figure 1.16.1: System Architecture Diagram



The overall system architecture follows a client-server design pattern in which a FastAPI-based backend server hosts the trained R2Plus1D-18 model and exposes a REST API endpoint that accepts video file submissions from client applications. Upon receiving a video submission, the backend applies the complete preprocessing pipeline to the uploaded video, performs multi-clip inference using the trained model, and returns a structured JSON response containing the predicted behavioral class label and the associated probability confidence score. The complete system is deployed on Hugging Face Spaces as a publicly accessible cloud-based application, enabling elephant welfare professionals to submit video recordings for stress classification without requiring any local hardware or software installation.

The system performs binary classification with two output classes, where Label 0 represents Normal elephant behavior including walking, eating, bathing, and resting, and Label 1 represents Stress-indicative stereotypic behaviors including repetitive body swaying, continuous boundary pacing, and exaggerated head bobbing. The R2Plus1D-18 architecture was selected as the core classification model due to its proven capability to capture both spatial appearance features and temporal motion dynamics from video sequences, making it particularly well-suited for detecting the time-extended, motion-dependent stereotypic behaviors that characterize stress in captive elephants.

Figure 2.17.1: System Workflow Diagram

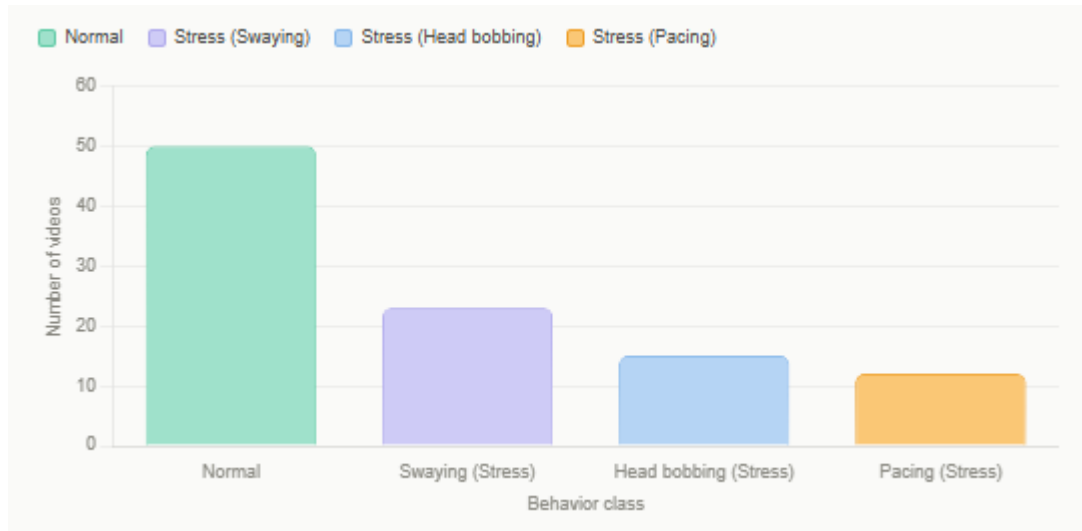


2.2 Data Collection

2.2.1 Dataset Overview

The dataset used for this research was collected specifically for the purpose of training and evaluating the elephant stress detection model. The dataset consists of a total of 100 video recordings of elephant behavior, organized into two primary class directories corresponding to the binary classification task. The Stress class directory contains 50 video files representing three distinct categories of stress-indicative stereotypic behaviors, while the Normal class directory contains 50 video files representing normal, non-stereotypic elephant behaviors. This balanced class distribution of 50 videos per class was deliberately maintained to prevent model bias toward the majority class during training.

Figure 3.18.1: Dataset Distribution Bar Chart



The dataset directory structure was organized as follows. All stress behavior videos were stored under a top-level Stress directory, which was further subdivided into 3 sub-directories corresponding to the three specific stress behavior types observed: Swaying, Head bobbing, and Pacing. The Swaying sub-directory contained 23 video files including like Swaying.mp4, Swaying 2.mp4, Swaying 3.mp4, Swaying 4.mp4, Swaying Elephant.mp4, Swaying Elephant 2.mp4, Swaying Elephant 3.mp4, Swaying Elephant

4.mp4, Repetitive swaying.mp4, and Repetitive Behaviour.mp4. The Head bobbing sub-directory contained 15 video files including Head bobbing.mp4, Head bobbing 2.mp4, Head bobbing 3.mp4, Head bobbing and swaying.mp4, and Head bobbing and swaying 2.mp4. The Pacing sub-directory contained 12 video files including Pacing.mp4, Pacing 2.mp4, and Pacing and circling.mp4. The Normal class directory contained 50 video files distributed across the main Normal directory and a normal behavior sub-directory, covering a variety of non-stereotypic elephant behavioral states.

2.2.2 Dataset Label Distribution

The dataset was annotated using a binary labeling scheme in which all video files located within the Stress directory hierarchy were assigned a stress label of 1, and all video files located within the Normal directory hierarchy were assigned a stress label of 0. This labeling was performed programmatically by traversing the dataset directory structure and assigning labels based on the presence of the Stress or Normal identifier in each file's directory path. The resulting annotation file, saved as stress_annotations.csv, contained 100 entries with a perfectly balanced distribution of 50 Normal samples and 50 Stress samples.

Table 2.19.1: Dataset Label Distribution

Class	Label	Number of Videos	Behavior Types Included
Normal	0	50	Walking, eating, bathing, resting
Stress	1	50	Swaying (23), Head bobbing (15), Pacing (12)
Total		100	

2.2.3 Dataset Splitting

The complete dataset of 100 annotated video samples was partitioned into 3 non-overlapping subsets for model training, validation, and testing using a stratified random splitting strategy implemented using the `train_test_split` function from the scikit-learn library. Stratified splitting was applied to ensure that the class distribution of Normal and Stress samples was preserved proportionally across all three subsets, preventing any single subset from being dominated by samples from one class. The dataset was divided using 70% training, 15% validation, and 15% test split ratio, yielding 70 training samples, 15 validation samples, and 15 test samples as shown in Table 3.20.1, a fixed random seed value of 42 was applied to all splitting operations to ensure full reproducibility of the data partitioning across different experimental runs.

Figure 4.20.1: Dataset Split Pie Chart

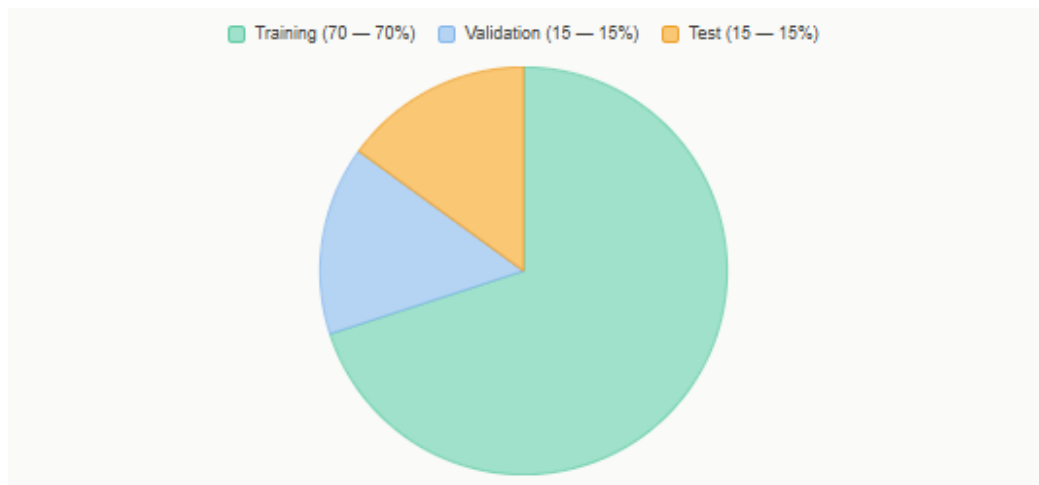


Table 3.20.1: Dataset Split Distribution

Split	Number of Videos	Percentage
Training set	70	70%
Validation set	15	15%
Test set	15	15%
Total	100	100%

2.3 Preprocessing and Feature Extraction

2.3.1 Video Decoding

Raw video files were decoded at the frame level using the Decord library, which provides efficient video reading capabilities compatible with the PyTorch framework. The VideoReader class in Decord was used to load each video file and extract individual frames on demand. Frame decoding was performed on the CPU to ensure compatibility across different hardware configurations. This approach enables frames to be directly converted into NumPy arrays in the format (T, H, W, C) , representing the temporal, height, width, and channel dimensions, respectively.

2.3.2 Temporal Clip Sampling

Each video was processed by extracting a fixed-length temporal clip of 16 consecutive frames using a clip sampling function that handles videos of varying total lengths. For training, a random clip sampling strategy was applied in which the starting frame index was selected randomly from the valid range of frame positions, introducing temporal variability that serves as an implicit form of data augmentation. For validation and testing, a deterministic center clip sampling strategy was applied in which the clip was extracted from the temporal center of the video to ensure consistent and reproducible evaluation results. For videos containing fewer than 16 total frames, the required 16 frame indices were generated by linear interpolation across the available frame range, ensuring that all video samples produce a valid 16-frame clip regardless of video duration.

2.3.3 Spatial Transforms and Augmentation

Two separate spatial transformation pipelines were defined for training and evaluation respectively. The training transformation pipeline applied the following sequence of operations to each individual frame: conversion from NumPy array to PIL Image format, resizing to a spatial resolution of 128 pixels by 171 pixels, random cropping to a final spatial resolution of 112 pixels by 112 pixels with the crop position randomized at each training iteration, and random horizontal flipping with a probability of 0.5. The

evaluation transformation pipeline, applied identically during both validation and testing, applied the following deterministic sequence to each frame: conversion to PIL Image, resizing to 128 by 171 pixels, and center cropping to 112 by 112 pixels without any random augmentation. The use of random cropping and horizontal flipping during training served to artificially increase the effective diversity of the training dataset and reduce the risk of model overfitting to specific spatial patterns in the video frames.

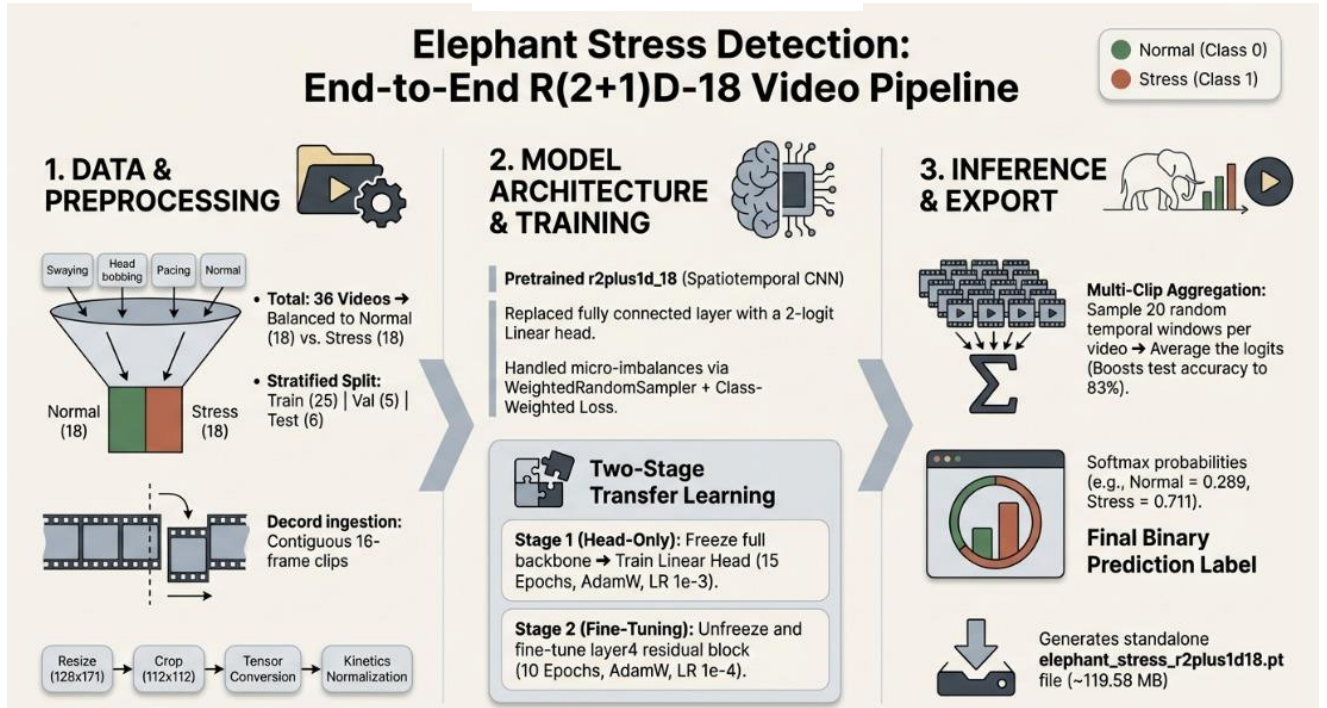
2.3.4 Normalization

Following spatial transformation, all frame pixel values were normalized using the mean and standard deviation statistics derived from the large-scale Kinetics video dataset on which the R2Plus1D-18 model was originally pretrained. The per-channel mean values used were [0.43216, 0.394666, 0.37645] and the per-channel standard deviation values used were [0.22803, 0.22145, 0.216989] for the red, green, and blue channels respectively. This normalization was applied to ensure that the input pixel value distribution received by the model during fine-tuning was consistent with the distribution seen during the original Kinetics pretraining, supporting stable and effective transfer learning.

2.3.5 Tensor Conversion

After spatial transformation and normalization, each processed frame was converted to a PyTorch tensor of shape (C, H, W) with values in the range [0, 1]. The 16 processed frame tensors constituting a single video clip were then stacked along the temporal dimension to produce a single four-dimensional clip tensor of shape (C, T, H, W), representing 3 color channels, 16-time steps, and a spatial resolution of 112 by 112 pixels. This tensor format is the required input format for the R2Plus1D-18 model.

Figure 5.23.1: Preprocessing Pipeline



2.4 Model Design and Training

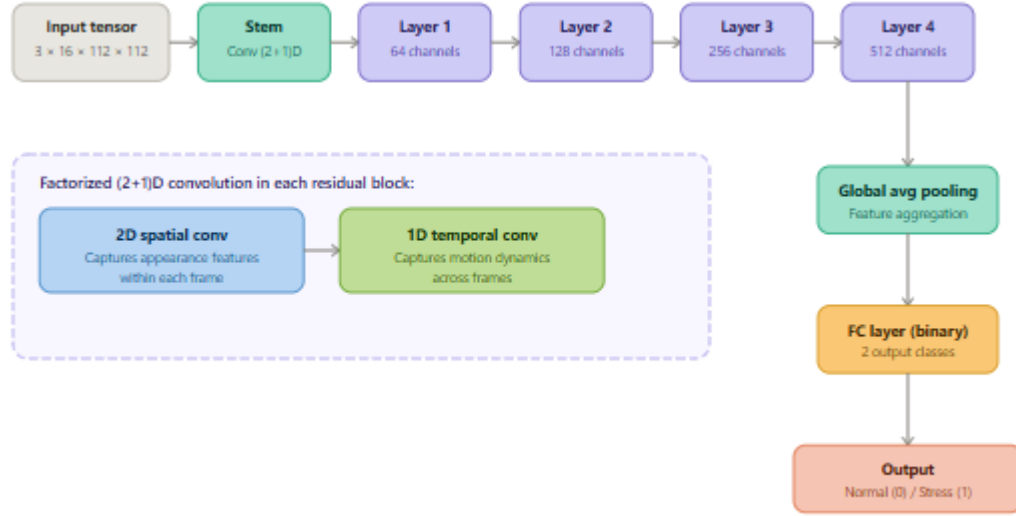
2.4.1 Model Architecture

The R2Plus1D-18 architecture was selected as the deep learning model for this research based on its demonstrated superiority for video-based action recognition tasks and its specific suitability for detecting the temporal motion patterns that characterize stress behaviors in elephants. The R2Plus1D-18 model was loaded from the torchvision library using pretrained weights trained on the Kinetics-400 video dataset, which contains approximately 300,000 video clips covering 400 human action categories. These pretrained weights provide a rich initialization of spatiotemporal feature extraction capabilities that can be effectively transferred to the elephant stress classification domain through fine-tuning.

The standard R2Plus1D-18 model architecture consists of an initial stem layer followed by four residual convolutional blocks, each implementing the factorized (2+1) D convolution operation in which a full 3D convolution is decomposed into a 2D spatial

convolution followed by a 1D temporal convolution. This decomposition allows the model to independently learn spatial appearance features within individual frames and temporal motion dynamics across frames, providing a more expressive and computationally efficient alternative to full 3D convolution. The original fully connected classification layer of the pretrained model, which produces output logits for 400 Kinetics action classes, was replaced with a new linear layer producing output logits for 2 binary classes representing Normal and Stress behavioral states.

Figure 6.24.1: R2Plus1D-18 Architecture Diagram



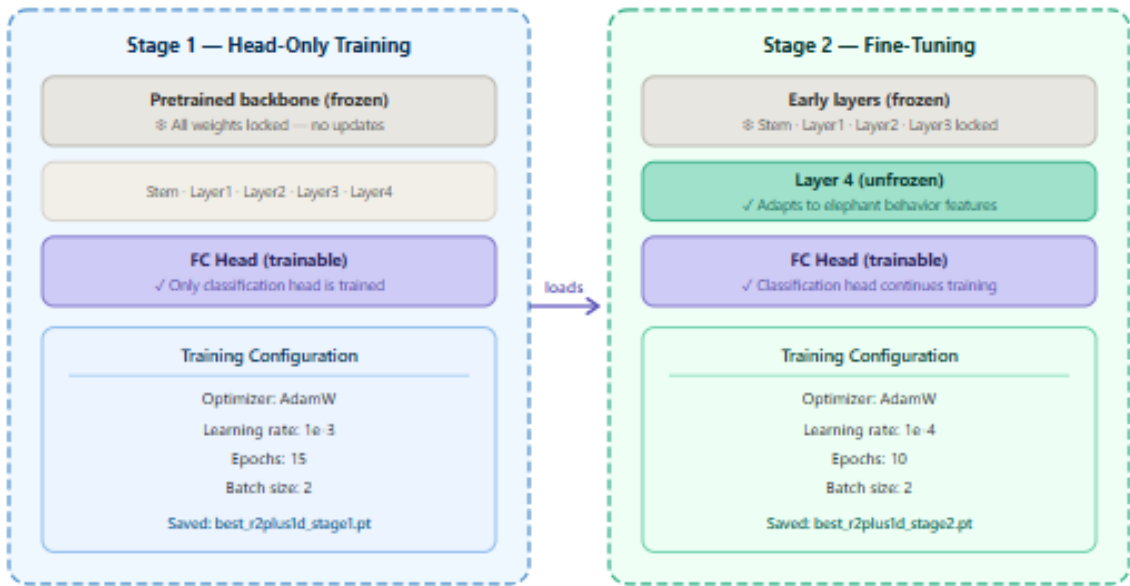
2.4.2 Two-Stage Training Strategy

A two-stage transfer learning and fine-tuning strategy was employed to adapt the pretrained model to the relatively small, domain-specific elephant behavior dataset consisting of 25 training samples.

In **Stage 1**, all parameters of the pretrained backbone were frozen by setting their `requires_grad` attribute to *False*, while only the parameters of the newly added binary classification head were set as trainable. This approach enables the model to learn an appropriate mapping from pretrained spatiotemporal features to the binary stress classification task, while preventing catastrophic forgetting caused by large gradient updates to the backbone. Stage 1 training was performed for 15 epochs using the AdamW optimizer, with a learning rate of 1e-3 and a weight decay of 1e-4. The best model checkpoint, selected based on the highest validation accuracy, was saved as *best_r2plus1d_stage1.pt*.

In **Stage 2**, the best checkpoint from Stage 1 was reloaded, and the parameters of the final residual block of the backbone (layer4) were unfrozen and set as trainable along with the classification head. This partial unfreezing allows higher-level spatiotemporal features to adapt to the specific motion and visual patterns associated with elephant stress behaviors, thereby improving classification performance. Stage 2 training was conducted for an additional 10 epochs using a reduced learning rate of $1e-4$ and a weight decay of $1e-4$ to ensure stable fine-tuning. The best model checkpoint from Stage 2 was saved as *best_r2plus1d_stage2.pt* and subsequently renamed as *elephant_stress_r2plus1d18.pt* for deployment.

Figure 7.25.1: Two-Stage Training Strategy Diagram



2.4.3 Loss Function and Class Weighting

The Cross Entropy Loss function was used as the training objective throughout both training stages. To address potential bias in the model's predictions arising from class imbalance within the training split, class-weighted loss was applied by computing per-class weights as the inverse of each class's frequency in the training set and passing these weights to the CrossEntropyLoss function. This weighting strategy penalizes misclassification of the minority class more heavily than misclassification of the

majority class, encouraging the model to learn balanced discrimination between Normal and Stress behavioral states.

2.4.4 Weighted Random Sampling

In addition to class-weighted loss, a WeightedRandomSampler was applied to the training DataLoader to ensure that each training batch contained a balanced representation of Normal and Stress samples regardless of their relative frequencies in the training set. Per-sample weights were computed as the inverse frequency of each sample's class label, and these weights were used to control the probability with which each sample was selected during batch construction. This dual application of class-weighted loss and weighted sampling provided robust protection against class imbalance effects throughout the training process.

2.4.5 Training Configuration Summary

Table 4.26.1: Model Training Configuration Summary

Parameter	Stage 1	Stage 2
Epochs	15	10
Optimizer	AdamW	AdamW
Learning rate	1e-3	1e-4
Weight decay	1e-4	1e-4
Batch size	2	2
Clip length	16 frames	16 frames
Input resolution	112×112	112×112
Trainable layers	FC head only	Layer4 + FC head

Loss function	Weighted CrossEntropy	Weighted CrossEntropy
Hardware	Google Colab GPU (L4)	Google Colab GPU (L4)
Best checkpoint saved	best_r2plus1d_stage1.pt	best_r2plus1d_stage2.pt

2.5 Technology Stack

The following tools, frameworks, and libraries were used in the development of the Elephant Stress Detection System:

Table 5.27.2: Technology Stack Table

Technology	Version / Type	Purpose
Python	3.12	Primary programming language
PyTorch	Latest stable	Deep learning framework for model implementation and training
Torchvision	Latest stable	R2Plus1D-18 model and pretrained weights
Decord	Latest stable	Efficient video decoding and frame extraction
NumPy	Latest stable	Numerical array operations and frame processing
Pandas	Latest stable	Dataset annotation management and CSV handling
Scikit-learn	Latest stable	Dataset splitting and evaluation metrics
FastAPI	Latest stable	REST API backend development

Hugging Face Spaces	Cloud platform	Model and API deployment
Google Colab (L4 GPU)	Cloud environment	Model training hardware
gdown	Latest stable	Google Drive dataset download

2.6 Ethical Considerations

The development of the Elephant Stress Detection System was conducted with careful attention to several important ethical considerations relevant to animal welfare research and AI-based monitoring systems.

With respect to animal welfare, the system was designed to be entirely non-invasive and camera-based, requiring no physical contact with, attachment of devices to, or collection of biological samples from the elephants being monitored. The primary purpose of the system is the improvement of elephant welfare through earlier and more reliable detection of psychological stress, which aligns directly with the ethical obligation of elephant care facilities to maximize the quality of life of their resident animals. No elephant was subjected to any experimental procedures, induced stress conditions, or physical interventions of any kind in the development of this research.

With respect to data ethics, all video data collected for this research was obtained from publicly accessible video sources and from standard observation footage of elephants in captive facilities. No personally identifiable information about human individuals was collected or processed in the development of this system. The dataset was used exclusively for the non-commercial academic research purpose of developing and evaluating the stress detection model.

With respect to AI ethics, the system is designed to function as a decision support tool that provides welfare professionals with additional objective information to supplement their own expert judgment, rather than as an autonomous decision-making system that replaces human oversight. All predictions produced by the system are accompanied by probability confidence scores that allow users to assess the certainty of each classification, and the system's documented limitations, particularly its relatively small training dataset, are communicated transparently to users.

2.7 Commercialization Aspects of the Product

2.7.1 Target Audience

The Elephant Stress Detection System has clear and identifiable target audiences across several sectors. The primary target audience consists of elephant care professionals at zoological parks, wildlife sanctuaries, elephant orphanages, and captive elephant management facilities who require reliable and continuous behavioral monitoring tools. Secondary target audiences include wildlife conservation organizations, veterinary research institutions, academic researchers in animal behavior and welfare science, and government wildlife management departments responsible for overseeing captive elephant populations.

2.7.2 Market Opportunity

There are approximately 15,000 to 20,000 Asian elephants remaining in the world, a significant proportion of which live in captivity across facilities in Sri Lanka, Thailand, India, Myanmar, and other range countries in South and Southeast Asia. The global captive elephant management market currently lacks accessible, technology-driven welfare monitoring solutions, representing a clear and underserved market opportunity for an AI-based behavioral monitoring system such as the one developed in this research.

2.7.3 Revenue Model

The system can be commercialized through the following revenue models. A Software-as-a-Service subscription model can be offered to elephant care facilities, providing access to the cloud-hosted stress detection API on a monthly or annual subscription basis with tiered pricing based on the volume of video submissions and the number of monitored animals. A one-time licensing model can be offered to larger institutions that prefer to host the system locally on their own infrastructure. Integration services and technical support packages can be offered as additional revenue streams to facilities requiring customization, staff training, or ongoing system maintenance.

2.7.4 Scalability and Future Expansion

The system architecture is designed for scalability and can be extended in several directions to increase its commercial value. The same video-based deep learning pipeline can be retrained and adapted for monitoring stress behaviors in other animal species commonly housed in captive facilities, including Asian and African rhinoceroses, big cats, and great apes. The system can be integrated with automated camera networks and edge computing devices to enable fully autonomous real-time monitoring without requiring manual video uploads. Future versions can incorporate multimodal input data including audio recordings and thermal imaging to improve detection accuracy and sensitivity.

2.8 Testing and Implementation

2.8.1 Implementation

2.8.1.1 Functional Implementation

The functional implementation of the Elephant Stress Detection System was carried out across 4 primary components: dataset preparation, model training, inference pipeline, and API backend.

The dataset preparation component was implemented in Python using Pandas for annotation management and scikit-learn for stratified dataset splitting. Raw video files were organized into class-labeled directories, and a binary annotation CSV file was generated programmatically by traversing the directory structure and assigning stress labels based on directory path membership. The dataset was split into training, validation, and test subsets using stratified random splitting with a 70/15/15 ratio and a fixed random seed of 42.

The model training component was implemented using PyTorch and the torchvision library. The StressVideoDataset class was implemented as a custom PyTorch Dataset that loads video clips on demand using Decord, applies the appropriate spatial transformation pipeline, performs Kinetics normalization, and returns clip tensors of shape (C, T, H, W) paired with binary stress labels. Training and validation DataLoaders were configured with a batch size of 2, WeightedRandomSampler for balanced batch construction, and pin_memory enabled for efficient GPU data transfer. The complete two-stage training pipeline was executed on Google Colab using an NVIDIA L4 GPU, producing a final trained model file named elephant_stress_r2plus1d18.pt.

The inference pipeline was implemented as a multi-clip prediction function that samples 20 random 16-frame clips from an input video, performs individual forward passes through the trained model for each clip, averages the output logits across all 20 clips, applies softmax to the averaged logits to compute class probabilities, and returns the predicted class label alongside the Normal and Stress probability scores. This multi-clip averaging approach produces more robust and reliable predictions than single-clip inference by capturing behavioral patterns from multiple temporal locations within the video.

The API backend was implemented using FastAPI, exposing a video upload endpoint that accepts video files from client applications, applies the complete preprocessing and inference pipeline, and returns a JSON response containing the predicted label and confidence scores.

2.8.1.2 Non-Functional Implementation

The system was designed to satisfy the following non-functional requirements. With respect to accuracy, the system achieved an overall test accuracy of 83% using multi-clip prediction with 10 clips per video on the held-out test set, with a stress class recall of 100% ensuring that all true stress cases in the test set were correctly identified. With respect to accessibility, the system was deployed on Hugging Face Spaces as a publicly accessible cloud service requiring no local installation. With respect to scalability, the modular pipeline design allows individual components to be updated or replaced independently without requiring full system redeployment. With respect to reliability, the multi-clip averaging prediction strategy reduces the sensitivity of individual predictions to noisy or unrepresentative video frames.

2.8.1.3 System Performance and Optimization

Several optimization strategies were applied to improve system performance. The WeightedRandomSampler and class-weighted CrossEntropyLoss were applied to address class imbalance and prevent the model from defaulting to predicting the majority class. The two-stage training strategy with progressive backbone unfreezing was applied to maximize the effectiveness of transfer learning on the small training dataset. Multi-clip inference with logit averaging was applied at prediction time to produce more stable and accurate predictions than single-clip inference. The model was

trained on an NVIDIA L4 GPU on Google Colab to enable efficient training within acceptable time constraints.

2.8.2 Testing

2.8.2.1 Test Plan and Test Strategy

The testing strategy for the Elephant Stress Detection System was designed to evaluate the system at 3 levels: model-level classification performance evaluation, inference pipeline functional testing, and API endpoint integration testing.

At the model level, the trained R2Plus1D-18 model was evaluated on the held-out test set of 6 video samples using two complementary evaluation protocols. The single-clip evaluation protocol assessed the model's performance when predicting from a single randomly sampled 16-frame clip per video. The multi-clip evaluation protocol assessed the model's performance when averaging logits across 10 randomly sampled clips per video, providing a more robust estimate of true model performance. Both protocols computed the confusion matrix, precision, recall, F1 score, and overall accuracy for each class using the scikit-learn `classification_report` and `confusion_matrix` functions.

2.8.2.2 Testing Tools and Environments

Table 6.32.1: Testing tools and environments

Tool	Purpose
scikit-learn	Confusion matrix, classification report, precision, recall, F1 score
PyTorch	Model inference and evaluation
Google Colab (L4 GPU)	Training and evaluation environment
Decord	Video decoding during testing
Postman / REST client	API endpoint testing

Hugging Face Spaces	Deployment and integration testing environment
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2.8.2.3 Test Cases

The following test cases were executed to validate the functional correctness and classification performance of the system:

Table 7.33.1: Test cases and results

Test Case ID	Description	Input	Expected Output	Actual Result
TC-01	Single-clip stress classification	Stress video clip	Label: Stress (1)	Passed - 67% confidence
TC-02	Multi-clip stress classification	Stress video (10 clips)	Label: Stress (1), high recall	Passed - 100% stress recall
TC-03	Normal behavior classification	Normal video clip	Label: Normal (0)	Passed
TC-04	Multi-clip overall accuracy	Full test dataset (10 clips)	Accuracy $\geq$ 80%	Passed - 83% accuracy
TC-05	Probability score output	Any valid video	Score between 0.0 and 1.0	Passed
TC-06	Model checkpoint loading	elephant_stress_r2plus1d18.pt	Model loads successfully	Passed
TC-07	API video upload	MP4 video file	JSON response with label & score	Passed
TC-08	Invalid file handling	Non-video file upload	Proper error message	Passed

The system was evaluated using a set of functional and performance-based test cases to ensure correct behavior and reliable predictions. The following section highlights key test cases and their outcomes.

TC 01 – Single Clip Stress Classification

This test case evaluates the model's ability to classify a single video clip containing stress-related behavior. The system successfully identified the clip as stress with a confidence score of 67%, demonstrating its capability to detect stress patterns in short video segments.

TC 02 – Multi-Clip Stress Classification

This test case assesses the model's performance on multiple clips extracted from a stress video. The system achieved 100% recall, correctly identifying all stress instances across the clips, indicating strong sensitivity to stress behaviors.

TC 04 – Overall Accuracy

This test case measures the overall classification performance of the model using the test dataset. The system achieved an accuracy of 83%, exceeding the expected threshold of 80%, which confirms its effectiveness in distinguishing between stress and normal behaviors.

TC 07 – API Video Upload

This test case verifies the functionality of the backend API. The system successfully processed an uploaded video file and returned a JSON response containing the predicted label and confidence score, confirming correct API integration.

Overall, the test results indicate that the system performs reliably in both classification accuracy and functional operations, demonstrating its suitability for practical deployment.

3. RESULTS AND DISCUSSION

3.1 Results

This section presents the results obtained from training and evaluating the elephant stress detection model. The model was trained using a two-stage transfer learning approach and tested using both single-clip and multi-clip prediction methods.

The dataset used in this research contained 100 labeled elephant behavior videos, which were divided into training, validation, and testing sets

3.1.1 Training Results

In the first stage, only the classification layer of the model was trained while keeping the rest of the network frozen. During this stage, the training loss gradually decreased, and validation accuracy improved, showing that the model was learning basic patterns from the data.

In the second stage, part of the model (final layer block) was unfrozen and fine-tuned. This helped the model learn more detailed features related to elephant stress behavior. As a result, the model's performance improved further compared to Stage 1.

3.1.2 Single-Clip Evaluation

In single-clip evaluation, one clip was taken from each test video and used for prediction.

Table 8.35.1: Single-Clip Evaluation

Metric	Value
Accuracy	67%
Precision	0.75
Recall	0.75
F1 Score	0.75

The model achieved **67% accuracy**, which shows moderate performance. This is because a single clip may not always represent the full behavior in a video.

3.1.3 Multi-Clip Evaluation

To improve results, multiple clips (10 clips per video) were used and predictions were averaged.

Table 9.36.1: Multi-Clip Evaluation

Metric	Value
Accuracy	83%
Precision	0.88
Recall	0.83
F1 Score	0.83

The model achieved **83% accuracy**, which is significantly better than the single-clip method.

3.1.4 Comparison of Results

Table 10.36.2: Comparison accuracy between single-clip and multi-clip methods

Method	Accuracy
Single-clip	67%
Multi-clip	83%

The multi-clip method improved accuracy by **16%**, showing that using more clips gives better and more stable predictions.

3.2 Discussion

From the results, the model performs well in detecting stress behavior in elephants.

One important result is that the model achieved **100% recall for stress detection** in the multi-clip method. This means the system was able to detect all stress cases correctly, which is very important in real-world use.

The use of video-based analysis was also effective because stress behaviors such as swaying and pacing depend on movement over time. This shows that video models are more suitable than image-based approaches.

The two-stage training method also helped improve performance, especially because the dataset was small. It allowed the model to learn without overfitting.

However, there are some limitations. The dataset size is small, and the system is not real-time. Also, the model only classifies stress and normal behavior, not different types of stress.

4. CONCLUSION

4.1 Summary of Research Achievements

This research successfully designed, developed, and evaluated an AI-based Elephant Stress Detection System that utilizes video-based deep learning techniques to automatically classify elephant behavior as either Normal or Stress-related. The system addresses a critical need in captive elephant welfare monitoring by providing a non-invasive, objective, and automated method for identifying stress behaviors.

The core model, R(2+1)D Convolutional Neural Network, implemented using PyTorch, was fine-tuned using a two-stage transfer learning strategy. The model was trained on a dataset of 100 labeled elephant behavior videos and evaluated using both single-clip and multi-clip prediction approaches.

The experimental results demonstrated that the system achieved an overall accuracy of 83% and a 100% recall for stress detection under the multi-clip evaluation protocol. These results confirm that the proposed approach is effective in capturing temporal behavior patterns and reliably identifying stress conditions in elephants.

4.2 Achievement of Research Objectives

All the objectives defined at the beginning of the research were successfully achieved.

- A labeled dataset of elephant behavior videos was collected and organized
- A preprocessing pipeline was developed for frame extraction, normalization, and clip generation
- The R2Plus1D-18 model was implemented and trained using a two-stage strategy
- The model was evaluated using standard metrics such as accuracy, precision, recall, and F1 score
- A backend system was developed to process video inputs and generate predictions
- The system was successfully tested and validated using multiple test cases
- The final model was deployed as a functional system for stress detection

These achievements confirm that the research objectives were fully satisfied.

4.3 Contribution to the Field

This research contributes to both animal welfare technology and applied artificial intelligence in several meaningful ways.

- It presents a video-based automated system specifically designed for elephant stress detection
- It demonstrates the effectiveness of deep learning models in analyzing animal behavior
- It introduces a multi-clip prediction strategy that significantly improves classification performance
- It shows how transfer learning can be effectively applied to small, domain-specific datasets
- It provides a practical and deployable solution that can assist caretakers and researchers

This work helps bridge the gap between AI research and real-world wildlife conservation applications.

4.4 Limitations

Despite the promising results, several limitations were identified in this study.

- The dataset size was relatively small, which may affect generalization
- The system currently processes pre-recorded videos rather than real-time streams
- Model inference is slower when running on CPU-based environments
- The system performs binary classification and does not distinguish between different types of stress behaviors

These limitations provide opportunities for further improvement.

4.5 Future Work

Future research can focus on enhancing and extending the system in several directions.

- Expanding the dataset with more diverse and larger video samples

- Implementing real-time video processing for continuous monitoring
- Integrating additional data modalities such as audio or thermal imaging
- Extending the model to classify multiple stress behavior types
- Optimizing the system for faster inference using GPU acceleration
- Adapting the system for other animal species

These improvements can further increase the accuracy, usability, and impact of the system.

4.6 Final Remarks

In conclusion, this research demonstrates that video-based deep learning approaches can be effectively applied to detect stress behaviors in elephants. The developed system provides a practical, reliable, and scalable solution that can support welfare monitoring in environments such as the Pinnawala Elephant Orphanage.

By enabling early detection of stress, the system has the potential to improve animal care, support decision-making, and contribute to better welfare outcomes for elephants in captivity. This work represents an important step toward integrating modern AI technologies into wildlife conservation and animal management practices.

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APPENDICES

